

Machine Learning Workshop 2025
Expressive Text to Speech
November 27th 2025 v1

Due date: Thursday December 18th 2025 23.59h.

Goals:

- This is an exploratory workshop to study some of the state-of-the-art (SOTA) Text-to-Speech (TTS) engines that are capable of voice cloning and/or expressing emotions.
- Gain hands-on experience with deploying a TTS engine, get insight in its speech quality, complexity and computational efficiency.
- Obtain an understanding of the workings of SOTA TTS architectures at a conceptual level.
- Write a scientific technical note on 1 selected recent TTS Engine capable of voice cloning and/or expressing emotions.

Note: The workshop should not take much more than 3 - 4 hours to finish. If you encounter any problems when trying to deploy the selected TTS engine on your system, take a pragmatic route.

Grading: 0-10

Introduction

Current TTS engines are designed to not only produce natural intelligible speech, but also allow voice-selection or -personalization and are able to produce convincing intonation and emotional expression. Furthermore, speech should be generated efficiently and with low latency. Tacotron [1] and its successors were one of the first end-to-end TTS engines producing natural sounding speech, and could be trained and even fine-tuned with different characteristic voices. Recently, several TTS engines are capable of expressive speech and even reasonably convincing voice cloning using a single short recording of a reference voice. See the following links for some examples:

Links:

1. Elevenlabs: <https://elevenlabs.io> (commercial)
2. Higgs Audio V2 [2]: <https://github.com/boson-ai/higgs-audio>
(can do Voice-Cloning at a reasonable quality level: [voice clone](#))
3. Chatterbox [3]: <https://github.com/resemble-ai/chatterbox> ([voice clone](#))
4. Coqui TTS: <https://github.com/coqui-ai/TTS>
(NB Tacotron is available as one of the many TTS engines of this TTS Toolbox.)

2) has been tested on Ubuntu 24.04.3 LTS, and Python 3.12.3 virtual-env.

3) and 4) have been tested on Ubuntu 24.04.3 LTS, and Python 3.10.19 virtual-env.

Evaluation Text

Use the following text to evaluate the performance of your selected TTS Engine(s):

"He found himself standing in a landscape that looked exactly like a giant chessboard. On every black square there was a monster: there were two-tongued snakes and lions with three rows of teeth, and four-headed dogs and five-headed demon kings and so on. He was, so to speak, looking out through the eyes of the young hero of the story. It was like being in the passenger seat of an automobile: all he had to do was watch, while the hero dispatched one monster after another and advanced up the chessboard towards the white stone tower at the end."

Assignment:

- a) Use a version of Tacotron as a baseline to produce natural speech for the Evaluation Text (see above). Evaluate the result on efficiency, latency, naturalness, intelligibility, and expressiveness. Save the natural speech result as a wav-file *tacotron.wav* and add it to your .zip file. NB If you encounter problems deploying Tacotron, you can use another (older) TTS engine. Clearly mention which and why.
- b) Find a very recent TTS engine that is able to produce emotional speech and/or Voice Cloning. Use your selected TTS engine to produce natural speech for the Evaluation Text, and evaluate the result on efficiency, latency, naturalness, intelligibility, and expressiveness. Save the natural speech result as a wav-file *<name_of_your_selected_tts_engine>.natural.wav* and add it to your .zip file.
- c) Evaluate the emotional expressiveness of your selected TTS engine by using your own text and your own selected emotional states (for example: neutral, sad, angry, excited, etc., but you are free in selecting any emotional or expressive state). Save the respective speech-results as .wav files and add them to your .zip file. If your selected TTS engine allows it, do a similar evaluation for voice cloning.

Report your findings in the form of a so-called technical note. One A4 font size 11, with Title, short abstract, short introduction, your TTS engine selection with a short paragraph on its concepts, a paragraph on its ease of deployment and performance, a paragraph with your experimental results (in one table), conclusions, references.

Submit your technical note as a single .pdf file together with a results.zip file containing all your .wav results on Brightspace before Thursday December 18th 2025 23.59h.

References and Links:

- [1] Y. Wang, R. Skerry-Ryan, D. Stanton, Y. Wu, R.J. Weiss, N. Jaitly, Z. Yang, Y. Xiao, Z. Chen, S. Bengio, et al., Tacotron: Towards end-to-end speech synthesis, 2017, arXiv preprint arXiv:1703.10135.
- [2] Boson AI, Higgs Audio V2: Redefining Expressiveness in Audio Generation, 2025
GitHub repository: <https://github.com/boson-ai/higgs-audio>,
Release blog available at <https://www.boson.ai/blog/higgs-audio-v2>
- [3] Resemble AI, Chatterbox-TTS, 2025.
GitHub repository: <https://github.com/resemble-ai/chatterbox>